

MAUKE, Cook Islands

WMO: 918400

Lat: 20.137S

Lon: 157.344W

Elev: 9

StdP: 101.22

Time Zone: -10.00 (W10)

Period: 82-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.5	19.4	12.3	8.9	21.3	13.1	9.4	21.5	10.6	22.6	9.5	22.4	1.2	130	0.309

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.5	30.7	26.3	30.3	26.1	29.8	25.9	27.1	29.6	26.8	29.2	26.5	28.9	3.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	21.9	28.5	26.1	21.5	28.4	25.9	21.2	28.2	85.8	29.8	84.4	29.5	82.8	28.8	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	8.1	6.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	25.2	26.8	27.3	27.4	26.4	25.2	24.1	23.5	23.3	23.6	24.3	25.3	25.8	(6)
(7)		DBStd	1.91	0.96	0.96	0.94	1.23	1.24	1.28	1.64	1.53	1.46	1.36	1.16	1.16	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	5563	520	484	539	492	472	423	419	412	408	443	459	491	(10)
(11)		CDD18.3	2521	262	251	281	242	214	173	161	153	158	185	209	233	(11)
(12)		CDH23.3	17786	2414	2517	2779	2077	1362	756	614	534	609	921	1394	1808	(12)
(13)		CDH26.7	2869	457	620	684	391	125	41	33	23	35	69	151	240	(13)
(14)	Wind	WSAvg	3.2	3.1	2.8	2.7	3.0	3.4	3.2	3.2	3.5	3.1	3.5	3.5	(14)	
(15)	Precipitation	PrecAvg	1647	226	200	184	179	154	73	86	82	72	99	113	180	(15)
(16)		PrecMax	2183	786	586	505	355	496	215	310	291	235	233	271	450	(16)
(17)		PrecMin	682	10	28	25	34	23	24	15	21	20	8	18	31	(17)
(18)		PrecStd	344	160	126	112	76	98	47	63	55	51	56	67	96	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.7	31.3	31.5	30.7	29.4	28.5	28.5	28.2	28.4	29.0	29.8	29.7	(19)
(20)			MCWB	26.4	26.5	26.5	26.5	25.5	24.2	24.6	24.3	24.0	24.5	25.6	25.4	(20)
(21)		2%	DB	30.1	30.7	30.8	30.1	28.7	27.7	27.6	27.4	27.6	28.2	28.9	29.3	(21)
(22)			MCWB	26.2	26.3	26.4	26.1	25.1	23.9	24.0	23.7	23.8	24.1	24.9	25.1	(22)
(23)		5%	DB	29.6	30.2	30.2	29.5	28.1	27.0	26.9	26.7	26.9	27.5	28.2	28.7	(23)
(24)			MCWB	25.9	26.1	26.2	25.6	24.7	23.4	23.5	23.3	23.3	23.7	24.4	24.8	(24)
(25)		10%	DB	29.0	29.7	29.7	28.8	27.4	26.4	26.1	25.9	26.1	26.8	27.6	28.1	(25)
(26)			MCWB	25.6	25.8	25.9	25.2	24.3	23.0	23.0	23.0	22.8	23.3	24.0	24.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	27.6	27.5	27.3	26.2	25.3	25.8	25.2	25.4	25.5	26.4	27.1	(27)
(28)			MADB	29.3	30.1	30.2	29.6	28.0	27.3	27.6	27.1	27.2	27.8	28.6	28.4	(28)
(29)		2%	WB	26.8	27.1	27.0	26.9	25.7	24.9	24.9	24.6	24.8	24.9	25.7	26.3	(29)
(30)			MADB	29.0	29.8	29.7	29.2	27.6	26.9	26.5	26.5	26.7	27.2	27.9	27.9	(30)
(31)		5%	WB	26.5	26.6	26.7	26.4	25.3	24.5	24.3	24.2	24.2	24.4	25.3	25.7	(31)
(32)			MADB	28.7	29.3	29.3	28.6	27.2	26.5	26.2	26.1	26.2	26.7	27.6	27.6	(32)
(33)		10%	WB	26.1	26.2	26.3	25.9	24.9	23.8	23.7	23.6	23.6	24.0	24.8	25.3	(33)
(34)			MADB	28.2	28.7	28.9	28.0	26.9	25.9	25.8	25.5	25.6	26.2	27.1	27.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.1	4.5	4.6	4.3	4.0	4.4	4.5	4.5	4.3	4.3	4.2	(35)	
(36)			MCDBR	4.9	5.2	5.3	5.0	4.5	5.0	5.2	5.3	5.3	4.8	4.8	4.7	(36)
(37)			MCWBR	2.3	2.4	2.3	2.3	2.5	2.7	2.9	2.9	2.6	2.6	2.5	2.5	(37)
(38)		5% WB	MADB	4.4	4.5	4.9	4.2	4.0	4.3	4.6	4.5	4.7	4.5	4.4	4.0	(38)
(39)			MCWBR	2.4	2.4	2.5	2.4	2.5	2.9	3.3	3.1	2.9	2.8	2.8	2.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.379	0.384	0.376	0.380	0.374	0.365	0.362	0.362	0.382	0.376	0.377	0.382	(40)	
(41)		taud	2.637	2.618	2.654	2.636	2.634	2.636	2.632	2.626	2.564	2.587	2.608	2.613	(41)	
(42)		Ebn at Noon	965	949	933	890	858	848	862	895	913	946	960	963	(42)	
(43)		Edh at Noon	101	101	94	90	84	80	83	89	101	103	103	103	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.29	6.13	5.62	4.78	3.98	3.79	3.98	4.69	5.59	6.07	6.40	6.29	(44)	
(45)		RadStd	0.51	0.46	0.48	0.31	0.33	0.21	0.23	0.30	0.28	0.51	0.47	0.49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page